

Project: Midterm

Data Analytics Midterm Project

Summary

Starting with raw data selection and progressing through Excel cleaning, visualization, and SQL analysis, this project simulates challenges you may face ahead in your new career. By working with real-world datasets, you will gain valuable experience in the full data analysis process and have an opportunity to see how far your skills have progressed.

The project emphasizes practical application of Excel and SQL skills to extract meaningful insights that could inform business decisions.

Your project will be evaluated by instructional staff primarily based upon the analysis report you provide. This is similar to the process you will experience with stakeholders in a professional environment.

Midterm Submission Page (<https://uprighted.instructure.com/courses/376/assignments/4412>)

Objectives:

- Demonstrate proficiency in selecting, importing, and documenting appropriate datasets for business analysis
- Apply data cleaning and transformation techniques to prepare raw data for analysis
- Create effective data visualizations and pivot tables that reveal patterns and trends
- Develop SQL queries of increasing complexity to answer specific business questions
- Practice documenting the entire data analysis process
- Communicate data-driven insights clearly through visualization, queries, and written reports

Materials:

Software

- **Microsoft Excel**
- **SQLite Studio**
- **Text Editor:** For documenting your process, findings, documenting SQL queries.
- **Instructions for creating a shareable google drive folder**
(<https://uprighted.instructure.com/courses/376/pages/g8c3624bb9db6a189ffd6306909e49d7b>)

Data

Below are 4 data sets that can be used for this project. In the case of a "cleaned" and "uncleaned" data set, you should use the "uncleaned" option:

1. **Data Science Job Posting on Glassdoor** ➞ (https://www.kaggle.com/datasets/rashikrahmanpritom/data-science-job-posting-on-glassdoor?select=Uncleaned_DS_jobs.csv)
2. **Audible Dataset** ➞ (https://www.kaggle.com/datasets/snehangsude/audible-dataset?select=audible_uncleaned.csv)
3. **Messy IMDB Dataset** ➞ (<https://www.kaggle.com/datasets/davidfuenteherraiz/messy-imdb-dataset>)
4. **Cafe Sales Dataset** ➞ (<https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training>)

For **10 extra points on your project**, you can use these data sets as a point of reference and **select your own data set** from Kaggle.

Note: If you select your own data set, it must be an **uncleaned set**. Selecting a cleaned set will significantly impact your grade.

Deliverables:

Google Drive folder that contains:

Excel Files:

- *ProjectName_Raw.xlsx: Your original, unmodified dataset*
- *ProjectName_Clean.xlsx: Your cleaned and transformed dataset*
- *ProjectName_Analysis.xlsx: Your visualizations and pivot tables*

SQL Files:

- *ProjectName_Queries.sql: Complete collection of all SQL queries*

SQL Analysis Report:

- *ProjectName_SQL_Analysis_Report.docx*

Tasks:

You will complete 9 total tasks that represent the progression of stages through the data analysis cycle.

For the first 4 tasks, you will work in Excel. After task 4, you will import your data and transition to SQL.

Excel

1. Dataset Selection and Import

1. Select Your Dataset:

- Choose a dataset that interests you and has potential for meaningful analysis.
- Document why you selected this dataset and what business questions you hope to answer.

2. Dataset Documentation:

- Column name
- Data type
- Brief description of what the data represents
- Potential business value of this data point
- Create a table that documents each column in your dataset:

3. Initial Data Assessment:

- Identify potential data quality issues (missing values, duplicates, inconsistent formatting).
- Document the data quality issues you'll need to address.
- Create a plan for cleaning each issue you've identified.

2. Data Cleaning and Transformation

Not all cleaning operations will be necessary for any given data set. This list is just for your reference to understand what may need to happen.

- **Handle Missing Values:** Identify missing data cells, select appropriate strategies (deletion/imputation), document your reasoning, and implement the chosen methods.
- **Fix Inconsistent Formatting:** Standardize text (case, spaces, special characters), ensure consistent date and numeric formats, and use appropriate Excel functions when needed.
- **Remove Duplicates:** Identify duplicate records, handle them appropriately, and document your process and rationale.
- **Document Your Process:** Maintain clear records of all data cleaning decisions, including strategies used and reasoning behind each approach.
- **Validate Results:** After cleaning, verify data integrity by checking that all issues have been resolved and that the cleaned dataset maintains its analytical value.

3. Pivot Table Analysis

1. Create Basic Pivot Tables:

- Create at least 3 different pivot tables that provide different views of your data.
- Each pivot table should answer a specific business question.

2. Pivot Table Analysis:

- The business question it answers
- The key insights revealed
- How this information could inform business decisions
- For each pivot table, write a paragraph explaining:

3. Combined Analysis:

- Create a worksheet that combines both visualizations and pivot tables.
- Show how the different analyses complement each other.
- Save your analysis as "ProjectName_Analysis.xlsx"

4. Data Visualization in Excel

1. Create Basic Visualizations:

- Create at least 3 different types of charts/graphs from your data.

2. Format Visualizations:

- Clear, descriptive title
- Properly labeled axes
- Appropriate color scheme
- Legend (if needed)
- Data labels or tooltips as appropriate
- Ensure each visualization has:

3. Visualization Analysis:

- What the visualization shows
- What business insights can be derived
- For each visualization, write a 2-3 sentence description of:

SQL

5. Data Import

1. Data Import:

- Import your cleaned data into your database tables
- Verify data integrity after import

2. Data Verification:

- Write queries to verify row counts match your original dataset
- Check for any data type issues or constraints that need addressing

6. Exploratory Query Analysis

1. Exploratory Queries: Develop at least 3 basic queries specifically designed to explore potential routes for analysis in your data.

- Count of records meeting specific criteria
- Sum or average of numeric fields with grouping
- Identification of minimum/maximum values
- Date-based filtering and analysis
- Text pattern matching using LIKE
- **Example Queries**

7. Business Query Analysis

1. **Business Insight Queries:** Develop at least 3 advanced queries specifically designed to answer key business questions related to your dataset.

- Time-series analysis (trends over time)
- Cohort analysis (grouping related records)
- Ranking and percentiles
- Running totals or moving averages
- Complex filtering with multiple conditions
- **Example Queries:**

8. Documentation and Insight Reporting

1. **Comprehensive Query Collection:**

- Compile all your queries into a single SQL file with clear comments
- Organize queries as either “Exploratory” or “Business”
- Include sample result sets for each query
- Save as "ProjectName_Queries.sql"

2. **Query Documentation:** Document your queries in preparation for the SQL Analysis Report described below.

- Describe your queries and Include sample result screenshots for each query
- Summary of the types of analysis performed
- Top significant insights discovered
- Paragraph explanation of how these insights could inform business decisions

9. Data Analysis Report

Comprehensive analysis report that demonstrates both your technical Excel & SQL skills and business analysis capabilities.

1. **Dataset Introduction:**

1. Source of the data
2. Size (number of tables and approximate records)
3. General subject matter (e.g., "retail sales data," "healthcare patient records")

2. **Analysis Objectives:**

1. Business questions being investigated
2. Patterns or trends the analysis hopes to uncover

3. Specific hypotheses being tested with the data

3. Analytical Approach and Findings:

1. Brief overview of methods used
2. Query explanations and insights

4. Business Recommendations:

1. Strategic suggestions based on your complete analysis
2. Actionable insights for stakeholders derived from your findings